mmWave Phase Interpolators

Architectures for phase control and generation for V/E Bands

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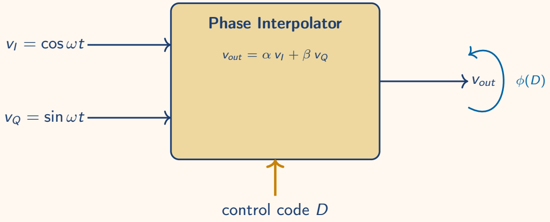
First Principles

What is a Phase Interpolator?

A circuit that produces a clock at any phase between two reference phases by taking a weighted sum.

- **Inputs:** two reference clocks, same frequency, known phase offset (typically 90° apart — I and Q)
- **Control:** digital code $\rightarrow$ analog weights (α, β)
- **Output:** one clock with phase $\phi \in [\phi_I, \phi_Q]$, same frequency

No frequency synthesis, no PLL — *just controlled mixing of two phases.*



Black box: 2 phases in, 1 phase out, code controls where.

UCI

Given two orthogonal clocks at ω:

$$v_I(t) = A\cos\omega t, \qquad v_Q(t) = A\sin\omega t.$$

Take a weighted sum with real weights α, β:

$$v_{out}(t) = \alpha\, v_I(t) + \beta\, v_Q(t) = A\big[\alpha\cos\omega t + \beta\sin\omega t\big].$$

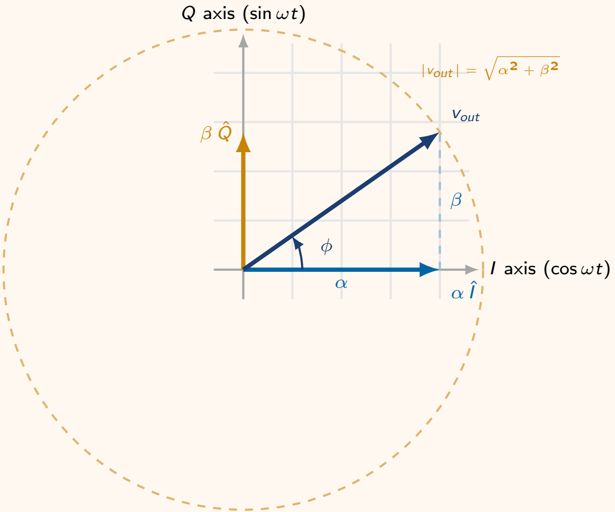
Apply the harmonic-addition identity $a\cos\theta + b\sin\theta = R\cos(\theta - \phi)$ with $R = \sqrt{a^2 + b^2}$, $\tan\phi = b/a$:

$$\boxed{v_{out}(t) = A\sqrt{\alpha^2 + \beta^2}\,\cos(\omega t - \phi)\,, \quad \phi = \arctan\frac{\beta}{\alpha}.}$$

Takeaway

The output *frequency* is unchanged. The output *phase* is set entirely by the weight ratio β/α — sweep the weights, sweep the phase from 0° (pure I) to 90° (pure Q).

I/Q Interpolation



Two knobs:

- Quadrant select (coarse) — pick two of $\{I, Q, \overline{I}, \overline{Q}\}$ for 360° coverage
- Weight ratio (fine) — N-bit DAC on β/α gives 2^N steps per quadrant

Phased-Array Beam Steering

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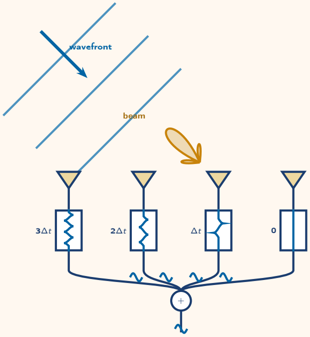

delays match TOA $\Rightarrow$ in phase at combiner
$\Delta t = \frac{d \sin\theta}{c} \;\Rightarrow\; \Delta\phi = \frac{2\pi d}{\lambda} \sin\theta$

Idea. Feed each antenna the *same signal* but with a progressive phase shift $\Delta\phi$ between neighbours. The array radiates a beam at angle

$$\boxed{\sin\theta = \frac{\lambda}{2\pi d}\,\Delta\phi.}$$

Change $\Delta\phi$ electronically $\rightarrow$ steer θ with no moving parts.

Why a PI per element?

- Continuous, monotonic phase
- Per-element skew calibration
- Same hardware across the band — only $\Delta\phi$ changes

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Array equation (d spacing, λ wavelength). Differentiate to map small $\Delta\phi$ to small $\Delta\theta$:

$$\Delta\phi = \tfrac{2\pi d}{\lambda}\sin\theta \implies \boxed{\Delta\theta = \frac{\lambda}{2\pi d\cos\theta}\,\Delta\phi_{\mathrm{min}}.}$$

For an N-bit PI ($\Delta\phi_{\mathrm{min}} = 2\pi/2^N$) at the typical half-wavelength spacing $d = \lambda/2$:
$\Delta\theta = (1/\cos\theta)\cdot 2^{-(N-1)}$ rad.

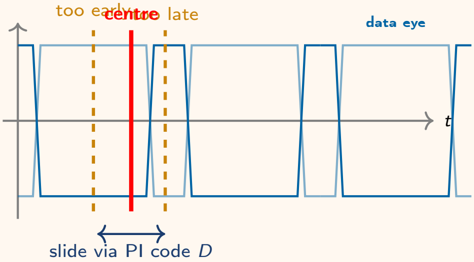
N	$\Delta\phi_{\mathrm{min}}$	$\Delta\theta_{@\,0^\circ}$	$\Delta\theta_{@\,60^\circ}$
4	22.5°	7.2°	14.3°
6	5.6°	1.8°	3.6°
8	1.4°	0.45°	0.90°

Wireline / SerDes

Problem. Receiver gets serial data with no clock. Local reference clock is at the right *frequency* but unknown *phase*.

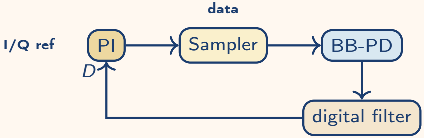
PI's job. Slide the sampling clock until it lands in the *centre of the data eye* — and keep it there as the link drifts.

$$\phi_{samp} = \frac{1}{2}(\phi_{rise} + \phi_{fall})$$



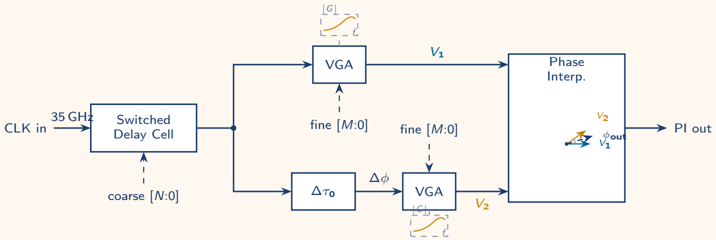
Why a PI?

Resolution of $360°/2^N$ at multi-GHz rates with no PLL in the loop. A bang-bang phase detector dithers the PI code ± 1 LSB at lock — the entire CDR loop is digital after the sampler.



Proposed Architecture

Proposed CMPI Architecture



Design parameters:

- $f_{in} = 35$ GHz, BW 30–40 GHz
- **Coarse** [N:0]: selects boundary phasors via switched delay
- **Fine** [M:0]: sets VGA weight ratio (α, β)
- $\Delta\tau_0$ fixes phasor gap $\Delta\phi = \omega\Delta\tau_0$
- VGA gain peaks toward band edge (broadband equalization)
- Output: $\phi = \arctan(\beta/\alpha)$

Questions?